The Water System Management Standard: Part 1 -- Asset Management

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Foreword

This document was developed by the mWater Foundation with funding in part by the United States Agency for International Development. mWater Foundation is an international non-governmental organization based in the United States that is dedicated to developing data-driven approaches for improving access to safe water and sanitation globally. The document draws on 10 years of experience by mWater in working with many government and civil society partners to develop digital tools and standardized approaches for managing data about water supply systems in low-resource areas of the world. For more information about how the Standard is implemented in the mWater digital platform, please visit: https://www.mwater.co/blog/asset-systems.

Introduction

This document was developed in response to demand from water authorities, utilities, government agencies, financial institutions, international development organizations, and engineering firms to specify a data structure and useful set of definitions to facilitate the capture, use, and sharing of data about water supply infrastructure.

Water supply infrastructure includes the physical assets and natural assets involved in the abstraction, storage, treatment, and distribution of water. Physical assets of a water system include hydraulic structures and components, electromechanical equipment, and civil engineering works. Natural assets, also known as green infrastructure, are natural landscapes and ecosystems that are strategically managed to provide ecosystem services that may enhance or substitute for physical assets in the production or management of water for human use.

Classification of assets

An asset management system tracks the key data about each asset that an organization considers valuable or important enough to actively manage. An organization must balance the need for data about assets with the effort involved in collecting data and keeping it up to date. There is no single right answer to the question of which assets need to be managed or tracked, since this depends on the needs and capabilities of the management organization. For this reason, this document does not mandate which asset types or attributes need to be used in an asset management system. To facilitate the development of an asset management system, this standard provides definitions for the most common types of assets found in water supply and distribution systems.

This standard classifies assets using a two-tiered structure that consists of **types** of assets that are organized into **classes**. The asset types are unique and sufficient to describe an asset. Classes are not required to define a type, they are simply useful groupings that reflect common industry practices. The standard currently recognizes 28 types of assets and 5 classes, as shown in Figure 1.

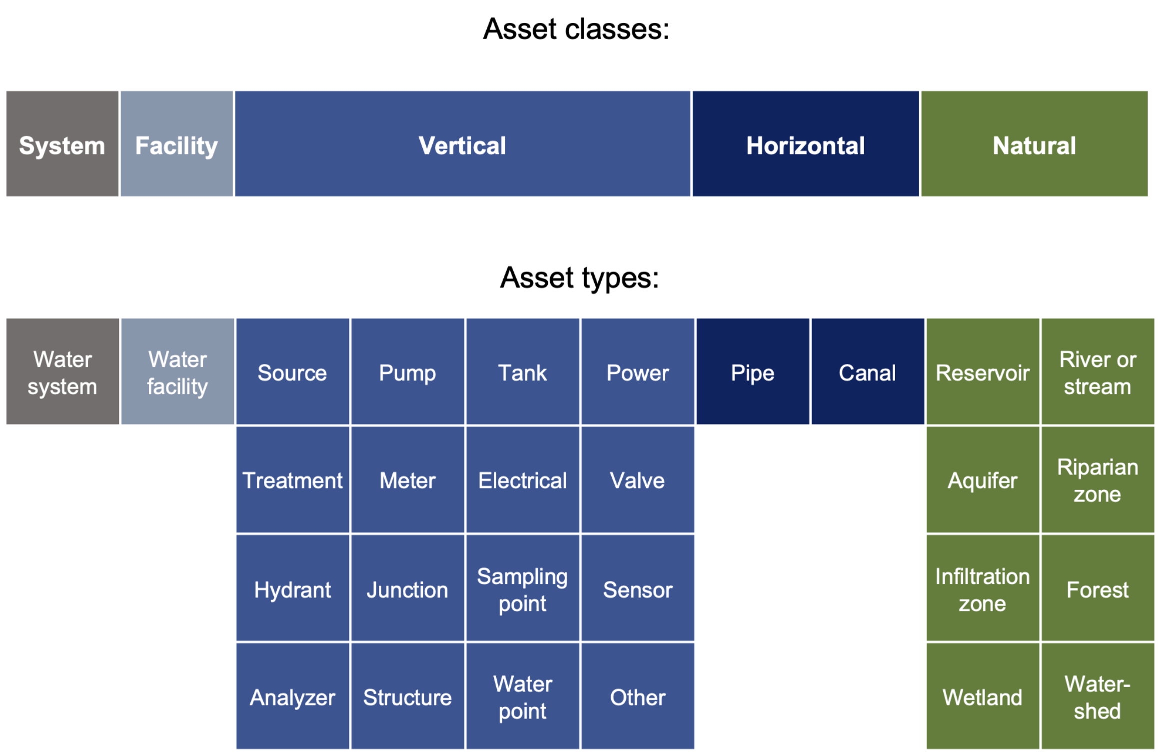

Figure 1: Asset classes and types

Attributes

Attributes are the individual pieces of data about a particular asset that describe important characteristics, such as the equipment manufacturer and model, condition and current operational status, and other technical details that relate to certain types of assets.

Any decision to include an attribute in a standard involves trade-offs between the need for comprehensive data and the practical challenges involved in collecting and updating the data. The following criteria were considered when evaluating potential attributes for this standard:

- **Is the attribute asset data?** Asset data pertains to the infrastructure and equipment used to provide services. Data on the quality of services provided, operations and maintenance, water quality, and customer service are all important for effective asset management, but these data are usually kept in other data systems.
- **Is the attribute in widespread use?** More consideration was given to defining attributes that are commonly used by water service providers, engineers, and technicians. This helps to determine the relative importance of collecting the data.
- **Is the attribute used to make decisions?** The focus of this standard is on data use by the service provider, as well as longer term data use for planning and financial modeling. Attributes that are needed to make specific decisions were emphasized over those that were more descriptive in nature.

The standard recognizes that there are some **general attributes** that apply to most or all assets and other **type-specific attributes** that only are relevant for certain kinds of assets. For example, most (but not all) types of equipment have a manufacturer, model, and serial number, but only boreholes have a casing diameter.

Distinguishing between general and type-specific attributes is helpful but not always sufficient to determine which kind of attribute applies to a certain asset, so the standard provides two additional concepts to help manage attributes:

- **Groups** are headings that organize data and keep similar attributes together. Every attribute in this standard has a group, even if there is only one attribute in the group. This is intended to provide consistency for programmers to design user interfaces that work the same way for any asset.
- **Conditions** are rules, based on the values of other attributes or the type of asset, that determine whether or not an attribute is applicable to a particular asset. For example, if the shape of a tank is cylindrical, then the diameter and height need to be recorded whereas a rectangular tank has a length and width but no diameter. Conditions may be applied to individual attributes or to groups of attributes.

The data model for attributes described in this section is summarized in Figure 2.

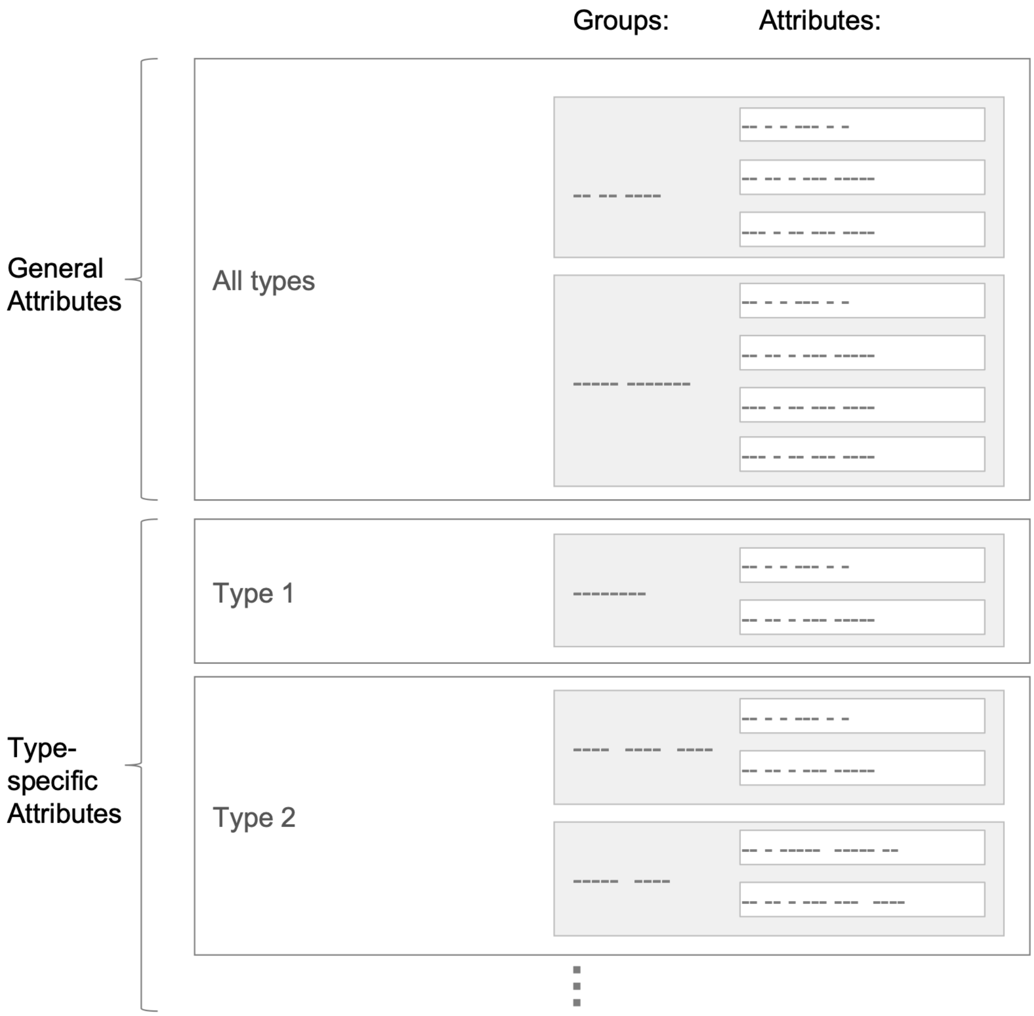

Figure 2: Illustration of general and type-specific attributes and groups of attributes

Organization of assets

Asset management systems typically use hierarchical classification schemes and parent-child relationships to help keep assets organized and help field staff to quickly identify or locate an asset. This standard accommodates up to 7 optional levels of hierarchical classification using attributes whose values are determined and enforced by the organization implementing the standard. In addition, any asset can be assigned a parent asset, creating a parent-child relationship between the two assets.

Standardized names and definitions

All types and attributes in this standard have unique names to improve clarity and interoperability in asset management systems. In addition, attributes have a unique numerical

ID code that specifies the class (1-digit), type (2-digits), and attribute (2-digits). In cases where attributes have a fixed set of possible values, the choices are also given names and are further defined by descriptions. The descriptions and definitions in this document are based on industry standard terms and cross-referenced against the American Water Works Association *Water Dictionary* (McTigue et at. 2010) and relevant ISO standards. Implementers are free to substitute their own terms or translations for the types, attributes, and choices specified in this standard but they should ensure that they are traceable back to the standard term for the purposes of data exchange and publication.

Using the standard

Organizations are encouraged to use this standard to design and develop infrastructure asset management systems that meet their unique needs, capabilities, and values. The standard is simple enough that it could be implemented using a spreadsheet or ledger, but organizations are encouraged to incorporate the standard into geographical information systems (GIS), online database systems, and mobile apps.

1 Scope

This document provides guidance for managing and sharing data about a water supply and distribution system.

This standard is applicable to any engineered or constructed works designed to deliver water for human use and consumption. Use of this standard does not require or depend on the use of a particular software application or database technology.

The scope of this document is limited to providing a standard description of the structure and types of data required to define the configuration, condition, and performance of the physical assets of a water supply system. Readers are encouraged to consult other standards and resources before implementing an infrastructure asset management process, including ISO standards 55000, 55001, and 55002, related to the asset management process, and ISO 24516 which provides specific guidance for water supply and wastewater system asset management.

2 Normative references

There are no normative references in this document.
A normative reference is an outside standard that is required in order to use or interpret the current document. In the current version of this standard, normative references are not necessary.

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.0.1 asset ID

a code used to identify a particular asset that is unique within the asset register (see Section 5.2 for requirements regarding unique identification of assets)

3.0.2 asset management system

the policies and procedures that an organization uses to manage assets

3.0.3 asset portfolio

assets that are within the scope of the asset management system

3.0.4 asset register

a complete listing of the asset information of an organization

3.0.5 attribute ID

a unique 5-digit code used in this standard to refer to a specific attribute

3.0.6 conduit

any artificial or natural duct, either open or closed, for conveying fluids

3.0.7 unit of measurement

a definite magnitude of a physical quantity, defined by convention or law, that is used as a standard for measurement of the same kind of quantity

4 Data types and units of measurement

Data types are used by computer programmers to help define the possible values a particular piece of data is allowed to have and the types of operations or transformations that can be performed on the data. This standard includes the data type for each attribute in Section 4.1 to help with the design of databases for asset data and to facilitate the transfer of asset data from one system to another. In cases where an attribute involves a physical measurement of some well-defined quantity, the standard also defines the units of measurement in Section 4.2, as well as the conversion factors between different units.

4.1 Data types

The following table describes the minimum data types needed to use this standard. In order to promote understanding among water professionals who might not be familiar with computer engineering terms, this section uses non-technical language to specify data types.

Table 1. Data types

Data type	**Description**
Text	Any combination of letters, numbers, punctuation, and spaces (also known as Character in some programming languages)
Choice	Text field that has a fixed set of possible values (also known as an Enumeration or Factor)
Number	Any integer or decimal number, consisting of a whole number part and an optional decimal part, with either a fixed or variable number of digits after the decimal point
Date	Calendar date according to the Gregorian calendar that includes the 4-digit year, 2-digit month, and 2-digit day in that order, and optionally separated by a delimiting character *Example: 2021.05.07 and 20210507 both specify the 7th day of May in the Year 2021*
Geometry	Location of a point, line, or polygon on the surface of the earth, according to the WGS84 geographic coordinate system (any common file format for geographical data may be used, including GeoJSON or ESRI Shape File)
Checkbox	Binary outcome that can only have values of TRUE or FALSE; checkbox questions default to FALSE unless the user changes the value to TRUE (also known as Boolean or Logical)
Image	Image file in any common format, such as JPG or PNG
Unit	Physical quantity that has both a magnitude (a number) and a unit of measurement

4.2 Units of measurement

Attributes that have the data type of Unit (see Section 4.1) include the name of the physical quantity that is being measured. The name “Unit” is followed by a colon (“:”) and then the name of the physical quantity. For example, the physical quantity of Length would appear in the Data type column of the attribute tables as: “Unit: Length”.

The units of measurement required to use this standard and their conversion factors are defined in Table 2. Each physical quantity has a base unit, which appears first in the table, and is defined by the International System of Units (SI). Conversion factors included in this section for SI units are derived from the SI Brochure (BIPM 2019).

SI units are preferable to other systems of measurement because they are widely adopted globally and avoid many of the errors included in traditional units. However, in engineering practice it is still common to find other units in use for certain applications. These traditional units are sometimes referred to as English units, inch-pound units, or United States customary units. For the purposes of this standard, these customary units are referred to as “US” units and their conversion factors are defined by the United States National Institute of Standards (NIST) Special Publication 1038 (Butcher et al. 2006).

The conversion factors are expressed with reference to the base unit, such that within each physical quantity the conversion factor for the base unit is always equal to 1. This means that a conversion factor can be treated as the ratio of the current unit to the base unit, or the amount of the base unit that would be equal to one current unit. For example, the conversion factor for centimeters, *0.01*, implies that *1 cm = 0.01 m*.

Any unit can be converted to any other unit by using the following equation:

$$Value\ in\ target\ unit\ = Value\ in\ current\ unit\ \times \frac{Current\ unit\ conversion\ factor}{Target\ unit\ conversion\ factor}$$

For example, to convert a length measurement of 10 centimeters into inches, multiply 10 by the conversion factor for centimeters (*0.01 m / cm*) and then divide by the conversion factor for inches (*0.0254 m / in*):

$$10\ cm\ \times\ \frac{0.01}{0.0254}\frac{m\ /\ cm}{m\ /\ in} = 3.94\ in$$

In the case of temperature, the conversion factor approach does not apply. Instead, follow the conversion formulas provided in Table 2.

Table 2. Units of measurement

Physical quantity / Symbol	**Name**	**Conversion factor**	**System**
Length			
m	meters	1	SI
cm	centimeters	0.01	SI
mm	millimeters	0.001	SI
µm	micrometers	1×10^{-6}	SI
nm	nanometers	1×10^{-9}	SI
km	kilometers	1000	SI
in	inches	0.0254	US
ft	feet	0.3048	US
yd	yards	0.9144	US
mi	miles	1609.344	US
Pressure			
Pa	pascals	1	SI
kPa	kilopascals	1000	SI
psi	pounds per square inch	6894.757	US
ft H_2O	feet of water (39.2 °F)	2988.98	US
in H_2O	inches of water (39.2 °F)	249.082	US
m H_2O	meters of water (4 °C)	9806.65	SI
cm H_2O	centimeters of water (4 °C)	98.0665	SI
mm Hg	millimeters of mercury (32 °F)	133.3224	SI
in Hg	inches of mercury (32 °F)	3386.38	US
atm	atmospheres	101325	SI
Flow rate			
m^3/s	cubic meters per second	1	SI
m^3/d	cubic meters per day	8.64×10^{4}	SI
L/s	liters per second	0.001	SI
L/d	liters per day	86.4	SI
ML/d	million liters per day	8.64×10^{7}	SI
gal/min (gpm)	gallons per minute	6.30902×10^{-5}	US
gal/d	gallons per day	4.38126×10^{-8}	US
Mgal/d (mgd)	million gallons per day	0.0438126	US

Voltage			
V	volts	1	SI
kV	kilovolts	1000	SI
MV	megavolts	1×10^6	SI
Power			
W	watts	1	SI
kW	kilowatts	1000	SI
MW	megawatts	1×10^6	SI
hp	horsepower	746	US
kVA	kilovolt-amps	1000	SI
Current			
A	amperes	1	SI
Volume			
m^3	cubic meters	1	SI
L	liters	0.001	SI
ft^3	cubic feet	0.02831685	US
gal	gallon	0.003785412	US
Mm^3	million cubic meters	1×10^6	SI
Mgal	million gallons	3785	US
acre-ft	acre-feet	1223.489	US
Time*			
d	days	1	SI
mo	months	30.436875	SI
yr	years	365.2425	SI
Temperature**			
°C	degrees celsius	$t_F = t_C \times 1.8 + 32$	SI
°F	degrees fahrenheit	$t_C = (t_F - 32) / 1.8$	US

** In this standard, time is used to refer to days, months, and years, where one year is defined using the average Gregorian year, defined as exactly 365.2425 days. Months are defined as 1/12 of a Gregorian year. This approximation is useful only for expressing approximate durations, such as intervals between scheduled maintenance activities, and is not intended to be used for calculating exact calendar dates.*

*** Temperature is a special case. Use the formula provided, where t_F = temperature in degrees Fahrenheit and t_C = temperature in degrees Celsius.*

5 Assets

5.1 Definition of an asset

An asset is an item, thing, or entity that has potential or actual value to an organization (ISO 55000). Water supply systems generally consist of physical assets, which are engineered, manufactured, or constructed; and natural assets, which are landscapes and ecosystems that perform some role in water supply, distribution, and treatment. For the purposes of database design, an asset consists of one single row in the table of assets.

5.2 Unique identification of assets

All assets shall be assigned an identification code, referred to as the Asset ID (section 6.4), that is unique with respect to all other assets included in the asset register. Identification codes may be text strings, numerical codes, or mixtures of text and numbers. In order to facilitate the tracking of maintenance interventions and failures over time, the asset ID shall be immutable, meaning it shall never be changed, even when an asset is moved or updated, and it shall never be reused when an asset is replaced or removed from service.

Organizational data managers and software providers should consider developing procedures or algorithms to check for and resolve duplicate asset IDs, especially when assets are created offline and later uploaded to a central database.

An organization may choose to use a hierarchical coding system, in which parts of the asset ID refer to types, facilities, or other categories that assist in identifying the location or function of the asset. Organizations setting up hierarchical coding systems should take into consideration the following:

- Assets are often moved from one location to another;
- Names of administrative regions and facilities change from time to time; and
- The asset owner or managing organization might change for a facility.

The asset coding system and staff procedures shall be designed to maintain a unique and immutable Asset ID over the entire service life of the asset.

5.3 Parent and child relationships

Any asset shall have an optional parent asset, which is defined by the value of the Parent ID attribute of the child asset. A child asset shall have only one parent asset, but parent assets may have an unlimited number of child assets.

5.4 Asset classes

Asset classes are useful groupings of asset types that fulfill similar functions and tend to have a common asset management strategy (AWWA 2018). In this document, asset classes are not necessary to include when describing an asset because the asset types (next section) are unique and sufficient to define the required attributes. However, the asset class can be useful for organizing lists of asset types during data collection, searching, and analysis.

The asset classes provided in the following table shall be accommodated in asset management systems.

Table 3. Asset classes

Asset class	Description	Geometry
System	Collection of assets that are managed in a similar manner to provide a service.	Point or polygon
Facility	Physical location where specific functions are performed.	Point or polygon
Vertical	Asset within a building or facility, or free-standing, often composed of multiple components; also known as an above ground asset (AWWA 2018).	Point
Horizontal	Asset which may be configured or networked for the purpose of moving materials or services from one place to another, such as pipes or *conduits* (AWWA 2018); also known as below ground assets, but may exist above or below ground.	Line
Natural	Strategically planned and managed natural lands, such as forests and wetlands, working landscapes, and other open places that provide associated benefits to human populations (Benedict and McMahon, 2006); also known as natural infrastructure.	Line or Polygon

5.5 Asset types

An asset type is a grouping of assets with common characteristics that distinguish them as a group (ISO 55000). In this document, asset types are used to determine which type-specific attributes can be added to that particular asset. The asset types described in the following tables shall be made available to include in the asset management system.

Table 4. Asset types

Asset class	#	Asset type	Description
System	101	Water system	Collection of assets that work together as a system to collect, treat, store, and deliver water to users
Facility	201	Water facility	Physical location where specific functions are performed in the production, treatment, and delivery of water; *the use of water facilities is optional and can be used to organize or locate the assets contained in large and complex treatment and distribution systems*
Vertical	301	Source	Location where a supply of water, such as a lake, river, spring, or groundwater aquifer, is abstracted for use in a distribution system
	302	Pump	Mechanical or electromechanical device that increases the hydraulic head (pressure) of a fluid flowing through it
	303	Tank	Structure or container used to hold water or other fluids
	304	Power	Device that supplies electrical power for use by electrical equipment, such as a generator, solar power system, wind turbine, or connection to a power grid
	305	Treatment	Equipment or process that improves the physical, chemical, biological, or aesthetic quality of water
	306	Meter	Device for measuring the Flow rate of water through a pipe
	307	Electrical	Any electrical or electronic device
	308	Valve	Mechanical device designed to close off or regulate the flow of water
	309	Hydrant	Device that provides the equipment needed to attach a fire hose to a water distribution network for use in extinguishing fires
	310	Junction	Section of a piped network that joins two or more pipes or *conduits*
	311	Sampling point	Location or access point in a water distribution network that allows for a water sample to be collected for testing
	312	Sensor	Device that measures a physical condition or parameter of interest
	313	Analyzer	Device that conducts periodic or continuous measurements of a water quality parameter

	314	Structure	Anything that is constructed or built with a fixed location on the ground
	315	Water point	Location or facility where users can access water
	399	Other vertical	Any other type of vertical asset not defined elsewhere in this standard
Horizontal	401	Pipe	*Conduit* that carries water or other fluids from one location to another
	402	Canal	Artificial open channel or waterway constructed for transporting water, connecting two or more bodies of water, or serving as a waterway
Natural	501	Reservoir	Impounded body of water or controlled lake in which water can be collected and stored
	502	River/stream	Large stream of water that serves as the natural drainage
	503	Aquifer	Geologic formation that is saturated and sufficiently permeable to transmit useful quantities of water to wells and springs
	504	Spring	Concentrated discharge of groundwater appearing at the ground surface
	505	Riparian zone	Banks or immediate border of a stream or river
	506	Infiltration basin	Shallow pond or depression used to infiltrate stormwater or runoff through permeable soils into a groundwater aquifer
	507	Forest	Land area characterized by woody vegetation greater than about 5 meters in height
	508	Wetland	Area that is inundated or saturated by surface water or groundwater at a frequency and duration sufficient to support a vegetation adapted for life in saturated soil conditions; includes swamps, marshes, and bogs
	509	Watershed	Area from which water drains and contributes to a particular stream or river

6 Attributes

6.1 Definition of an attribute

An attribute is a piece of data that describes a quality or characteristic of an individual asset.

6.2 Characteristics of attributes

Attributes and their usage shall conform to the following characteristics:

1) **Attributes whose values are unknown or not applicable shall be left blank.** The standard does not distinguish between a missing value and an unknown value, but implementers may choose to track additional metadata about attributes that are not answered, such as when an inspector was unable to determine the value.
2) **Attributes with conditions shall only be assigned a value when the condition is satisfied.** Some attributes are only applicable when another attribute has a certain value or when the asset has a specific class or type. These conditions, which may apply to individual attributes or groups of attributes, are provided in the attribute tables.

6.3 How to read and interpret attribute tables

All attributes defined in this standard are organized into tables that follow a consistent format with the following headings:

- **Code**: The 5-digit unique code assigned to each attribute.
- **Group**, Attribute, or *Choice* Name: The unique name used in the standard to refer to the attribute, group, or choice option. Note that the text formatting defines which of these three kinds of name is being defined:
  - **Group** names are in **bold font** and do not have an associated attribute ID.
  - Attribute names are in normal font and always have an attribute ID in the row where they appear.
  - *Choice* names are the names of the answer choices available for the attribute immediately above them and are displayed indented and in *italic* font.
- **Description**: A brief phrase or sentence to further define the meaning or use of the attribute or answer choice.
- **Data type**: The data type and unit of measurement defined (see Section 4) to be used with this attribute. In cases where more than one data type is permitted, types are separated by commas (,).

- **Conditions:** The condition that must all be met in order for the group or attribute to be applicable. Conditions are formatted with the name of the attribute, type, or class that the condition depends on, followed by a colon (:) and the values or answer choices that satisfy the condition. In cases where multiple values could satisfy the condition, each value is separated by a comma (,). Note that when a condition appears on the same row as a Group name, it applies to all attributes in the group (meaning all attributes that appear below the group name).

6.4 General attributes

The following attributes shall be available for all assets.

Table 5. General attributes for all assets

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Basic info			
00001	Type	Type of asset, according to the definitions in Section 5.5	Choice	
00002	Name	Name used by technical staff to refer to the asset; might be left blank for many lower level assets	Text	
00003	Asset ID	Unique code used to identify a particular asset in the asset register (see Section 5.2 for further details)	Text	
00004	Description	Additional information to help staff locate or identify an asset	Text	
00005	Photo	Photo of the asset	Image	
00006	Parent asset ID	Asset ID of the asset that is a parent to this asset (see Section 5.3 for further details)	Asset ID	
	Location			
00011	Location	Geographic location of the asset on the surface of the Earth; may be a point, line, or polygon	Geometry	
00012	Location precision	Estimated uncertainty in the geographic coordinates, expressed as a distance	Unit: Length	
00013	Elevation	Height above the surface of a defined ellipsoid model of the earth's surface, such as WGS84, or the local mean sea level; a consistent reference system should be used for all assets within a water system	Unit: Length	

00014	Elevation precision	Estimated uncertainty in the measurement of the elevation	Unit: Length	
00015	Administrative region	Name that describes the administrative geography of the location; multiple levels may be separated by commas starting with the most general area, for example: *Region, District, Municipality*	Text	
00016	Coverage area	Geographical area where the water system or service provides coverage or is legally obligated to serve	Geometry	Type: Water system
00017	Sector	Name for a sub-region of the coverage area that is defined by geography or the physical layout of the distribution network	Text	
00018	Route	Name or number of the meter reading route, which defines those meters that are assigned to a particular staff member or team	Text	Type: Meter, Water Point
00019	Address	Physical address or postal address where the asset is located; in the case of piped connections, the address of the customer	Text	
00020	Urban-rural classification	Degree of urbanization, as defined by national standards or the Degree of Urbanisation method (OECD 2021)	Choice	Type: Water system, Water point
	Urban	Cities or densely populated areas		
	Peri urban	Towns and semi-dense areas, suburbs, or peri urban areas surrounding cities		
	Rural	Regions with low population density		
00021	Community	Name of the community, town, or neighborhood where a water system or facility is located	Text	Type: Water system, water facility
00022	Accessible by road	Asset can be accessed by vehicles using a nearby road	Checkbox	
00023	Perimeter fence or wall installed	Fence or wall exists to provide physical security for the asset	Checkbox	
	Hierarchy	Hierarchy refers to a tiered set of names that help to locate or organize assets; the use of an asset hierarchy is optional and an organization may choose to use fewer levels than the 7 that are available in this standard		
00031	Level 1	Highest or most general level of categorization to which the asset belongs under the asset hierarchy	Text	

00032	Level 2	Second highest level of hierarchy	Text	
00033	Level 3	Third highest level of hierarchy	Text	
00034	Level 4	Fourth highest level of hierarchy	Text	
00035	Level 5	Fifth highest level of hierarchy	Text	
00036	Level 6	Sixth highest level of hierarchy	Text	
00037	Level 7	Lowest level of hierarchy	Text	
	Status and condition			
00041	Status	How the asset is currently being used or operated	Choice	
	Active	Asset is in use or ready to use		
	Backup	Asset is installed or available onsite to be used as a backup to other similar assets in case of maintenance or failure		
	Not functional	Asset is not currently operational due to a failure or maintenance problem		
	Under construction	Asset is not operational because it is under construction		
	Decommissioned	Asset is no longer in use because it is permanently out of service		
	Disposed	Asset has been sold, demolished, or relocated to a disposal site		
00042	Condition	The physical or structural integrity of the asset	Choice	
	Excellent	Designed to current standards, operable and well-maintained		
	Good	Showing some signs of minor wear and tear, with minimal impact on performance; or, operable and well-maintained but not designed to current standards		
	Fair	Structurally sound but some deterioration or damage that is beginning to impact performance or require more frequent maintenance		
	Poor	Functional but requires a high level of maintenance to continue operating or requires major rehabilitation or replacement		
	Very poor	Serious problems that impact performance or safety; major rehabilitation or replacement		

		required in the short term		
00043	Functional status	Most recent observation of whether a water system or water point is working as designed	Choice	Type: Water system, Water point
	Functional	In good working condition and regularly providing water according to the specifications in the original design		
	Partially functional	Provides water on a regular basis (possibly in a reduced capacity) but repairs are needed due to some maintenance issue or change in water quality or availability		
	Not functional	No longer providing water on a regular basis; this could be due to maintenance issues, changes in water availability or quality, or problems with physical access		
	Decommissioned	Permanently taken out of service, abandoned, or could not be found		
	Financial			
00051	Current replacement cost (USD)	Cost (in US Dollars) that would be incurred to replace the asset at the current time; this value requires regular updating based on depreciation and current market conditions; also known as CRC	Number	
00052	Expected useful life	Number of years that the asset is estimated to function after installation if it is well maintained; also known as EUL	Number	
00053	Probability of failure	Measure of the likelihood of a failure happening at any point in time, based on an analysis of the potential failure modes of the asset	Choice	
	Very low			
	Low			
	Moderate			
	High			
	Very high			
00054	Consequence of failure	Measure of the direct and indirect costs of asset failure, which may include financial, environmental, and societal costs	Choice	
	Very low			
	Low			

	Moderate			
	High			
	Very high			
00055	Asset risk	Business or organizational risk of asset failure, usually calculated from the probability and consequence of failure	Choice	
	Very low			
	Low			
	Moderate			
	High			
	Very high			
00056	Original cost (USD)	Total price paid (in US Dollars) to initially acquire an asset	Number	
00057	Current rehabilitation cost (USD)	Cost (in US Dollars) that would be incurred to rehabilitate the asset at the current time	Number	
	Technical			
00061	Manufacturer	Name of the business that produced or sold the asset	Text	
00062	Model	Manufacturer’s designation for the particular model of the asset	Text	
00063	Serial number	Unique identifying number or code assigned to a particular asset by the manufacturer or vendor, usually stamped or printed on the asset	Text	
00064	Additional technical details	Any other technical information about the asset that is not captured by the attributes of this standard	Text	
00065	Installation date	Calendar date when the asset was installed or purchased, which begins the period of useful life	Date	
00066	Installed by	Name of the company or organization that originally installed or purchased the system or facility	Text	Type: Water system, water facility
00067	Funded by	Name of the funding authority, lender, or donor who funded the purchase of the system or facility	Text	Type: Water system, water facility
00068	Rehabilitation date	Date of most recent work done to restore the asset to its original level of service and extend its life	Date	
00069	Last service date	Date of most recent service or maintenance performed	Date	
00070	Rehabilitated by	Name of the company or organization that performed the most recent rehabilitation of the asset	Text	Type: Water system, water facility

00071	Service interval	Recommended interval at which service or routine maintenance should be performed on the asset	Unit: Time	Type: Pump, tank, power, valve
	Hydraulic			Type: Pump, Tank, Treatment, Meter, Valve, Hydrant, Junction, Sampling point, Water point, Pipe
00091	Upstream asset	Asset ID of the upstream asset that supplies water to this asset	Asset ID	
00092	Nominal diameter	An approximate measurement of the internal diameter of a pipe, conduit, pump, or junction, which is used to classify according to standard sizes	Unit: Length	Type: Meter, Valve, Junction, Pipe
00093	Pipe length	Total length of pipe in a given section or run for use in hydraulic calculations and modeling	Unit: Length	Type: Pipe
00094	Material	Material of construction for pipes, junctions, valves, meters, and other liquid conduits	Choice	Type: Meter, Valve, Hydrant, Junction, Pipe
	Brass	Metal alloy of copper, zinc, and usually some lead		
	Bronze	Copper-base metal alloy that may or may not include tin, lead, or other metals in small amounts		
	Copper	Reddish metallic element that is ductile and malleable		
	Cast iron	Heavy brittle alloy of iron, carbon, and silicon that can be cast in specific shapes and has the ability to withstand shocks		
	Ductile iron	Similar material to cast iron, but with the primary graphite in a nodular or spheroidal form, giving it increased strength and flexibility		
	Stainless steel	Chromium alloy steel that is resistant to rusting and corrosion		
	Steel	Iron-base alloy containing up to 2 percent carbon that is heat treated		
	Galvanized steel	Steel that has a zinc coating to protect from corrosion and rust		
	PVC	Polyvinyl chloride, a type of polymer frequently used in pipes and containers		
	HDPE	High-density polyethylene, a type of polymer used in low-pressure applications		

	Plastic	Any high molecular weight polymer material that can be extruded, molded, or drawn into various shapes and sizes; use this category for any plastics that are not PVC or HDPE		
	Monel	Nickel alloy, primarily composed of nickel and copper, with small amounts of iron, manganese, carbon, and silicon, which is corrosion resistant		
	Other	Any material not otherwise defined in this section		
00095	Minor loss coefficient	Dimensionless coefficient that can be used to account for the head loss associated with turbulence in bends, junctions, meters, and valves	Number	Type: Junction, valve, meter
00096	Roughness coefficient	Dimensionless coefficient used to account for friction losses in flowing pipes in either the Hazen-Williams or Darcy-Weisbach head loss equation	Number	Type: Pipe
	Points of contact			Type: Water system, Water point
00171	Primary contact name	Individual or office responsible for management or administration of the water system or water point	Text	
00172	Primary contact position	Title or role in the organization of the primary contact	Text	
00173	Primary contact phone number	Phone number of the primary contact	Text	
00174	Secondary contact name	Additional individual who may be contacted if the primary contact is not available	Text	
00175	Secondary contact position	Title or role of the secondary contact	Text	
00176	Secondary contact phone number	Phone number of the secondary contact	Text	
	Metadata			
00181	Date updated	Calendar date of the most recent update made to any attribute	Date	
00182	Date inspected	Calendar date of the most recent inspection or audit of the asset, as defined by organizational asset management policies	Date	
00183	Alternate ID	Optional additional identification code or number that is used by other data systems or organizations to track the asset	Text	

00184	Import code	Identification code for an asset that was assigned by an external database or asset management registry, used to maintain backwards traceability for assets that have been imported	Text	

6.5 Type-specific attributes

The attributes defined in this section apply only to a specific asset type. Some asset types do not currently have type-specific attributes so they do not appear in this section.

6.5.1 System types

6.5.1.1 Water system

Table 6. Water system attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Water system info			Type: Water system
10201	Water system type	Categorization of the system as a distribution network or a point source	Choice	
	Distribution network	Water supply is distributed to multiple users via a network of pipes		
	Point source	Users access water at a single location		
10202	Network type	Categorization of piped networks as branching or grid style	Choice	Type of system: Distribution
	Branching	Pipes begin from a central point and split off into segments that are not interconnected to each other		
	Grid or mesh	Pipes split off into segments that are interconnected to each other downstream		
	Mixed	Network has aspects of both branching and grid styles		
10203	Management type	Type of organization that manages the operations, maintenance, and	Choice	

		finances of the water supply system		
	Utility	Organization created by government through legislative action or a private partnership or corporation with a legal mandate to provide water services to a designated area		
	Local government	Local or municipal government office or staff directly manage the system		
	Water user association	Community-based organization or cooperative group of water users (also known as water user committee, WASHCO, or community-based management)		
	Water authority	Government agency or regulatory body established for the purpose of managing water service provision		
	Private	Individual, partnership, or corporation that owns and operates a water supply system but does not have a legal mandate to provide services to all water users		
	Other	Any other management entity not described by the other choices		
	No management	No individual or organization can be identified that manages the system		
10204	Number of households in service area	Total number of households within the area that a service provider is legally mandated to offer services to	Number	
10205	Number of households currently served	Number of households with a functional connection or currently able to receive service (includes households with inactive, intermittent, or cut off services, if service could be re-established)	Number	
10206	System or network diagram	Photograph of a recent map or diagram showing the location of water supply and distribution infrastructure	Image	
10207	Total production of system	Amount of water that the system is able to supply, within current resource and design constraints	Unit: Flow rate	

6.5.2 Facility types

6.5.2.1 Water facility

Table 7. Water facility attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Water facility info			
20101	Water facility type		Choice	
	Intake	Facility located at the head of a *conduit* into which water is diverted		
	Well field	Group of wells treated as a single entity for administrative, production, and treatment purposes		
	Pumping station	Structure containing pumps and related piping, valves, and other equipment for pumping water		
	Distribution system	Network of *conduits* and related equipment by which water is distributed to consumers		
	Pipeline	Collection of pipes and related equipment that are jointed to provide a *conduit* through which fluids flow		
	Storage facility	One or more tanks or storage reservoirs and related equipment used to maintain pressure and store water to accommodate intermittent periods of higher usage		
	Treatment plant	Facility or area that contains various treatment works, process areas, and other equipment necessary to improve the quality of water		
	Process area	Area or building, often contained within a treatment plant, where a specific water treatment process is carried out		
	Other	Any other type of water facility not defined in this section		
20102	Process area type	Optional label used to categorize process areas	Text	Water facility type: Process area

6.5.3 Vertical types

6.5.3.1 Source

Table 8. Source attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Source info			
30101	Source type	Characterization of the source based on type of technology	Choice	
	Intake	Works or structures at the head of a *conduit* into which water is diverted		
	Spring	Concentrated discharge of groundwater appearing at the ground surface as a current of flowing water		
	Borehole	Deep hole with a small diameter that has been driven, bored, or drilled, in order to reach groundwater and constructed with casing or pipes to prevent collapse and contamination		
	Dug well	Shallow hole with a large diameter that is dug into the ground to reach groundwater		
	Rainwater catchment	System that collects rainwater from large surfaces, such as a roof or impermeable ground surface, so that it can be stored		
	Other	Any other type of source not otherwise defined in this section		
30102	Source meter installed	Meter is installed at the source to measure total flow	Checkbox	
30103	Water quality tested	Water quality testing has been performed at this source	Checkbox	
30104	Date of most recent sample	Calendar date of most recent water quality sampling at this source	Date	Water quality tested: True
30105	Name of water body	Name of the water body supplying the source, if one exists	Text	Source type: Intake
30106	Source pressure	Pressure, or head, of the water leaving the source	Unit: Pressure	Source type: Intake
30107	Nominal flow	Typical or rated volumetric water flow rate that the source provides when operating as designed	Unit: Flow rate	
30108	Minimum flow	Minimum Flow rate rate that the source provides during the driest period of the year	Unit: Flow rate	
30109	Maximum flow	Maximum Flow rate rate that the source can accommodate during	Unit: Flow rate	

		periods of high use or source water availability		
	Well data			Source type: Borehole, Dug well
30121	Well depth	Depth below the ground surface to which the well was drilled, driven, or dug; in boreholes, this may differ from the casing depth	Unit: Length	
30122	Well diameter	Inside diameter of the well; in boreholes, this is the drilled diameter, which will differ from the casing diameter	Unit: Length	
30123	Static water level	Water level below the ground measured when no water is being taken from the aquifer by pumping	Unit: Length	
30124	Dynamic water level	Water level below the ground measured when a pump is operating	Unit: Length	
30125	Groundwater temperature	Temperature of groundwater in the aquifer or discharged by a pump	Unit: Temp-erature	
30126	Pump or lifting device		Choice	
	Submersible pump	Electrically powered pump designed to fit inside the well casing and to operate below the water level		
	Hand pump	Human-powered mechanical pump that uses suction or positive displacement to manually lift groundwater to the surface		
	Rope and bucket	Rope attached to a bucket on one end and a structure or winch on the surface to manually lift groundwater		
	Other	Any other type of pump not otherwise defined in this section, including centrifugal or positive displacement surface pumps, wind, and animal powered pumps		
	None	No pump is currently installed on the well		
30127	Apron installed	Apron is installed to protect the well from back-contamination by surface water and to channel drainage away from the well	Checkbox	
30128	Monitoring well	Well is primarily used for monitoring water levels and/or water quality rather than for water production	Checkbox	
	Spring data			Source type: Spring
30141	Spring box present	Protective brick, masonry, or concrete structure is installed around	Checkbox	

		the spring so that water flows directly out through a pipe		
30142	Spring box length	Length of the longest side of the spring box, measured from the outside	Unit: Length	Spring box present: True
30143	Spring box width	Length of the shortest side of the spring box, measured from the outside	Unit: Length	Spring box present: True
30144	Spring box depth	Length of the vertical dimension of the spring box, measured from the outside	Unit: Length	Spring box present: True
30145	Overflow pipe present	Pipe is installed near the top of the spring box to allow water to exit when box is full	Checkbox	
30146	Drain pipe present	Pipe is installed near the bottom of the spring box to direct the flow from the spring	Checkbox	
	Borehole data			Source type: Borehole
30151	Casing diameter	Inside diameter of the well casing	Unit: Length	
30152	Depth of casing	Depth below the ground to which the casing is installed; below this depth there would normally be a screen installed	Unit: Length	
30153	Casing material	Well casing material	Choice	
	Steel	Iron-base alloy containing up to 2 percent carbon that is heat treated		
	Stainless steel	Chromium alloy steel that is resistant to rusting and corrosion		
	PVC	Polyvinyl chloride, a type of polymer frequently used in pipes and containers		
	Plastic	Any high molecular weight polymer material that can be extruded, molded, or drawn into various shapes and sizes; use this category for any plastics that are not PVC		
	Concrete	Mixture of cement and aggregate, usually reinforced with steel wire or rebar		
	Fiberglass	Reinforced plastic material composed of glass fibers embedded within a resin matrix		
	Galvanized steel	Steel that has a zinc coating to protect from corrosion and rust		
	Other	Any other type of casing material not described in this section		

30154	Screen length	Length of the well screen, which is a sleeve with slots, holes, gauze, or wire wrap installed below the casing to allow entry of water and prevent entry of sand or other large particles	Unit: Length	
30155	Pump position	Depth below ground to the top of a submersible pump	Unit: Length	Pump or lifting device: Submersible
30156	Photo of well log	One or more photographs of the well log, or drilling log, which usually contains a record of the soil and rock formations (lithology) encountered during the drilling of the well; may also contain data on the casing and backfill used to reinforce the well	Image	

6.5.3.2 Pump

Table 9. Pump attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Pump info			
30201	Pump type	Description of the pump mechanism or operating principle	Choice	
	Submersible	Electrically powered pump designed to fit inside the well casing and to operate below the water level		
	Axial flow	Fluid is accelerated in the direction of the flow axis by an impeller running inside a casing		
	Centrifugal	Fluid enters along the pump axis, is accelerated by an impeller, and exits at right angles to the shaft; also known as a radial flow pump		
	Reciprocating	Fluid is moved by one or more oscillating pistons or plungers		
	Positive displacement	Fluid is trapped and forced to move to the discharge side by a piston, gear, or rotary vane		
	Hand pump	Human-powered mechanical device that uses a suction or positive displacement pump, often connected to a lever via a rod or chain, to manually lift water		

	Hydraulic ram	Device that uses the water hammer effect to use some of the energy of inflowing water to lift a smaller volume of water		
	Other	Any other pump type not described in this section		
30202	Pump diameter	Maximum outer diameter of a submersible pump	Unit: Length	Pump type: Submersible
30203	Pump power rating	Power consumed by a pump operating under normal pressure and flow conditions, as defined by the manufacturer	Unit: Power	Pump type: NOT Hand pump
30204	Pump motor configuration	Whether the motor and pump should be tracked as a single asset or as separate assets; in cases where the motor is separate from pump, this generally means that motors are interchangeable between pumps of a similar design	Choice	Pump type: NOT Hand pump
	Integrated with pump	Motor and pump should be treated as the same asset		
	Separate from pump	Motor should be treated as a separate asset from the pump		
	Pump motor data			Pump motor configuration: Integrated with pump
30205	Pump motor voltage type	Type of electrical power required by the pump	Choice	
	Single phase AC	Alternating current electrical system that uses two wires: one neutral wire and one phase wire in which the voltage rises to a peak in one direction and then reverses to a peak in the opposite direction		
	Three phase AC	Alternating current electrical system that uses four wires: one neutral wire and three phase wires in which the voltage rises to a peak and reverses direction at different times		
	DC	Direct current system in which current always flows in one direction and the voltage does not change over time		
30206	Pump motor voltage	Normal operating voltage of the pump, as recommended by the manufacturer	Unit: Voltage	
30207	Pump motor max current	Current drawn by the pump when operating at the maximum or full	Unit: Current	

		load power rating		
30208	Pump motor max power	Maximum, or full load power rating of the pump, as recommended by the manufacturer	Unit: Power	
	Pump curve			
30221	Design flow	Flow rate of the pump required by the system design under normal operating conditions; should be within the acceptable flow range as defined by the pump curve	Unit: Flow rate	
30222	Design head	Head, or pressure, shown on the pump curve at the design flow rate	Unit: Pressure	
30223	Maximum flow (Q-max)	Maximum possible flow rate shown on the pump curve	Unit: Flow rate	
30224	Head at maximum flow (H-min)	Total head, or pressure, that corresponds to the maximum flow rate on the pump curve; also referred to as the minimum head	Unit: Pressure	
30225	Minimum flow (Q-min)	Minimum possible flow rate shown on the pump curve	Unit: Flow rate	
30226	Head at minimum flow (H-Max)	Total head, or pressure, that corresponds to the minimum flow rate on the pump curve; also referred to as the maximum head	Unit: Pressure	
30227	NPSH-R	Net positive suction head required (or minimum) by the suction side of the pump in order to prevent cavitation and evaporation (flashing)	Unit: Pressure	

6.5.3.3 Tank

Table 10. Tank attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Tank info			
30301	Tank type	Characterization of the tank design	Choice	
	Elevated	Located on and supported by a tower; also referred to as an overhead tank		
	Above ground	Constructed or permanently installed on top of the ground surface		
	Underground	Constructed or permanently installed partially or completely underground;		

		also referred to as an in-ground tank		
	Break pressure	Tank used for hydraulic isolation and surge protection installed in a pressurized system where the water surface is exposed to atmospheric pressure		
	Portable	Tank which can be moved from place to place; could be installed above or below ground		
	Other	Any other tank design not otherwise described in this section		
30302	Tank material	Main material of construction used in the tank	Choice	
	PVC	Polyvinyl chloride, a type of polymer frequently used in pipes and containers		
	Plastic	Any high molecular weight polymer material that can be extruded, molded, or drawn into various shapes and sizes; use this category for any plastics that are not PVC		
	Concrete	Mixture of cement and aggregate, usually reinforced with steel wire or rebar		
	Aluminum	Lightweight metal composed mainly of elemental aluminum that may be alloyed with other elements		
	Steel	Iron-base alloy containing up to 2 percent carbon that is heat treated		
	Composite	Structure that combines concrete with a steel support structure		
	Fiberglass	Reinforced plastic material composed of glass fibers embedded within a resin matrix		
	Masonry	Brick, stone, or concrete blocks that are bound together with mortar		
	Other	Any other type of tank construction not described in this section		
30303	Tank volume	Useful volume of the tank which can be supplied from the minimum to the maximum water levels.	Unit: Volume	
30304	Tank shape	Basic shape or form factor of the tank	Choice	
	Cylindrical	Circular base with vertical sides		
	Rectangular	Square or rectangular base with vertical sides		
	Other	Any other shape not otherwise		

		described in the section		
	Tank dimensions			
30311	Tank diameter	Diameter of the base of a cylindrical tank	Unit: Length	Tank shape: Cylindrical
30312	Tank height	Distance from the bottom to the top of the tank, measured from outside	Unit: Length	
30313	Tank length	Length of the longest side of the base of a rectangular tank	Unit: Length	Tank shape: Rectangular
30314	Tank width	Length of the shortest side of the base of a rectangular tank	Unit: Length	Tank shape: Rectangular
30315	Height of tank bottom	Distance from the ground level to the bottom of the inside of the tank; negative values are used for depths below the ground	Unit: Length	
30316	Minimum water level	Minimum water level allowed in the tank, measured from the tank bottom; this may be the height of the outlet or the minimum height allowed by a filling control	Unit: Length	
30317	Maximum water level	Maximum water level allowed in the tank, measured from the tank bottom; this may be the height of the overflow pipe or the maximum height allowed by a filling control	Unit: Length	
	Tank details			
30321	Filling control installed	Control is installed to ensure a desired range of tank fill levels	Checkbox	
30322	Type of filling control	Method or device used to maintain the water level in the tank by controlling the influent or effluent flow rate	Choice	Filling control installed: True
	Float valve	Flow into the tank is controlled by a gate or plug that is actuated by a float when the water level falls below a certain set point		
	Altitude valve	Automatically shuts off flow into an elevated tank when the water level reaches a predetermined level and automatically opens when the pressure in the distribution systems drops below the pressure in the tank		
	Other	Any other fill control method not described in this section		
30323	Fill level indicator installed	Device is installed that provides a visual or electronic measurement of the fill level	Checkbox	

6.5.3.4 Power

Table 11. Power attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Power info			
30401	Power type	Categorization of the power generating device based on design or principles of operation	Choice	
	Grid	Connection to an electrical transmission and distribution system		
	Generator	Diesel, gas, or petrol powered engine that converts rotary motion into electrical current		
	Solar	Solar or photovoltaic power system		
	Wind	Wind turbine or windmill equipped with an electrical generator		
	Other	Any other power type not defined in this section		
30402	Rated power output	Power output that can be sustained for long periods of time; for generators this is the continuous power rating	Unit: Power	
30403	Maximum power output	Power output that can be provided for short periods of time in response to surges in demand; for generators, sometimes referred to as the standby or intermittent power rating	Unit: Power	
30404	Nominal voltage	Voltage assigned to a system or circuit of a given voltage class for the purpose of convenient designation; the actual operating voltage may vary above or below this value	Unit: Voltage	
30405	Rated voltage	Maximum voltage at which the power system can be operated within thermal and safety limits	Unit: Voltage	
30406	Maximum current	Maximum safe current, or starting current, that can be supplied for a short period of time	Unit: Current	
30407	Type of current	Mode of transmission for electrical current (AC/DC)	Choice	
	Single phase AC	Alternating current electrical system		

		that uses two wires: one neutral wire and one phase wire in which the voltage rises to a peak in one direction and then reverses to a peak in the opposite direction		
	Three phase AC	Alternating current electrical system that uses four wires: one neutral wire and three phase wires in which the voltage rises to a peak and reverses direction at different times		
	DC	Direct current system in which current always flows in one direction and the voltage does not change over time		
30408	Frequency of current	Frequency at which the voltage rises to a peak, reverses, and rises to a peak again	Choice	Type of current: Single phase AC, Three phase AC
	60 Hz	60 cycles per second		
	50 Hz	50 cycles per second		
	Other	Any non-standard current frequency		
	Generator data			Power type: Generator
30421	Generator fuel	Type of fuel required by the generator	Choice	
	Gasoline or petrol	Fuel made from crude oil and other petroleum liquids that is often used in internal combustion engines		
	Diesel	Distillate fuel oil that is used in compression ignition engines		
	Other	Any other fuel used in generators that is not described in this section		
30422	Generator apparent power	Product of the voltage and current provided in an AC power system, which does not account for the reactive power required by inductive loads in the circuit; sometimes referred to by its most common unit of measurement: kVA	Unit: Power	Type of current: Single phase AC, Three phase AC
30423	Engine cooling system	Type of cooling system used by the generator engine	Choice	
	Air	Engine cooled by air circulation only		
	Liquid	Engine cooled by a circulating liquid, such as water or other coolant		
	Solar power data			Power type: Solar
30424	PV panel type	Photovoltaic panel type	Choice	
	Polycrystalline	Multicrystalline or poly panels, made of multiple pieces of silicon melted or		

		joined together		
	Monocrystalline	Single crystalline or mono panel, made from a single silicon crystal		
	Thin film	Thin, flexible panel made from amorphous silicon or other materials		
	Other	Any other type of photovoltaic panel type not otherwise described in this section		
30425	Total number of PV panels	Total number of photovoltaic panels installed in the power system	Number	

6.5.3.5 Treatment

Table 12. Treatment attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Treatment info			
30501	Treatment type	Categorization of treatment process by treatment method or objective	Choice	
	Screen or bar grate	Coarse screen or grate that removes large debris from water to prevent clogging of pipes and pumps		
	Filtration	Small diameter granular medium, such as sand, removes particulate matter from a liquid stream		
	Membrane filtration	Pressurized liquid is forced through a membrane to remove particulate matter		
	Reverse osmosis	Pressure-driven membrane removes ions, salts, and other dissolved solids and nonvolatile organics		
	Mixer	Agitation or blending process that creates a homogeneous mixture out of two or more components		
	Coagulator	Basin in which chemicals (coagulants) are added to destabilize the negative surface charges present in natural particles,		

		allowing them to accumulate		
	Flocculator	Device that uses gentle stirring to bring suspended particles together so they will form larger, more separable clumps called floc		
	Clarifier or sedimentation basin	Tank or basin in which water is retained to allow settleable matter, such as floc, settle by gravity		
	Chlorination	Device or basin that introduces chlorine, in liquid, solid, or gaseous form, for oxidation of organic matter or disinfection		
	Disinfection	Process that destroys or inactivates pathogenic organisms by physical or chemical means (use this category for any non-chlorine disinfection methods such as UV or ozone)		
	Aeration	Gas transfer process that causes a liquid stream to absorb a gas, such as air or oxygen		
	Air stripping	Passing air through liquid to remove volatile compounds		
	Desalinization	Process that uses distillation, reverse osmosis, or electrodialysis to remove dissolved mineral salts from seawater or brackish water		
	Solids disposal	Process to concentrate, dewater, thicken, or otherwise treat residual solids for safe disposal		
	Chemical feed	Addition of chemicals to process water to perform some treatment function, such as pH control or corrosion prevention		
	Other	Any other type of treatment process not already described in this section		
	Chlorination data			
30521	Chlorine type	Chemical form of chlorine used in a treatment process	Choice	
	Sodium hypochlorite (bleach)	Solution in which sodium hypochlorite (NaOCl) is hydrolyzed in water to form hypochlorous acid (HOCl); also known as liquid bleach		
	Calcium hypochlorite	Dry calcium hypochlorite ($Ca(OCl)_2$) in powder or pellets that can be dissolved in water, where it reacts to form hypochlorous acid (HOCl)		
	Chlorine gas	Gaseous molecular chlorine (Cl_2) that forms hypochlorous acid (HOCl)		

		and hypochlorite ion (OCl^-) when dissolved in water		
	Chloramine	Disinfectant produced by mixing chlorine and ammonia (NH_3); when added to water, forms mainly monochloramine (NH_2Cl) and dichloramine ($NHCl_2$)		
	Chlorine dioxide	Red-yellow gas that is very reactive and unstable; decomposes in water to form chlorite ion (ClO_2^-)		
30522	Dosing method	Method used to add chlorine to water to be treated	Choice	Chlorination type: Sodium hypochlorite (bleach), Calcium hypochlorite, Chloramine
	Dosing pump	Small pump draws concentrated chlorine solution from a tank and adds it to the water stream or tank		
	Drip feed	Constant head device, such as a float or bowl, located inside a tank feeds the concentrated chlorine solution at a flow rate that is independent of tank level		
	Constant head aspirator	Sealed container with a small outlet tube at the bottom and an air inlet tube at the top that provides a constant pressure and steady flow rate of chlorine solution		
	Gravity feed	Tank or bucket that uses gravity to feed a concentrated chlorine solution through a metering orifice or valve that controls the flow rate		
	Batch mixing	Chlorine bleach or powder is added to a large tank in batches		
	Tablet or cartridge	Water flows through a chamber or cartridge in which chlorine is slowly released from a solid material		
	Other	Any other chlorine dosing method not described in this section		

6.5.3.6 Meter

Table 13. Meter attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Meter info			
30601	Meter ID	Identification number of code used by the service provider to identify the meter	Text	
30602	Meter type	Type of metering performed: bulk metering of the distribution network versus domestic or commercial metering	Choice	
	Bulk	Used to measure flows throughout the distribution network to monitor performance and detect leaks		
	Domestic	Measures flow to a household or private connection		
	Commercial or industrial	Measures flow to a commercial or industrial user		
	Production	Measures flow from a source or treatment plant		
30603	Meter design	Design or operating principle of the meter	Choice	
	Turbine	Measures flow velocity based on the speed of a rotating turbine		
	Displacement	Captures definite volumes of water in a chamber with a piston or disk that discharges the water		
	Ultrasonic	Measures flow velocity by transmitting and receiving sound waves that pass diagonally through the flow path		
	Electromagnetic	Measures the induced voltage in a sensor that is generated by a fluid as it flows through a pipe		
	Jet	One or more jets of water impact on a turbine that measures flow rate		
	Other	Any other type of meter not otherwise described in this section		
30604	Minimum measurable flow rate	Lowest flow rate that can be measured by the meter	Unit: Flow rate	
30605	Meter measurement type	Type of measurement displayed by the meter (total volume, flow rate, or both)	Choice	
	Total volume	Totalizing meter, which displays only the total amount of volume that has flowed through the meter, often using a combination of a counter and a sweep hand		

	Flow rate	Instantaneous measurement of the current flow rate through the meter		
	Combined	Meter display includes both the instantaneous flow rate and the total volume that has flowed through the meter		

6.5.3.7 Electrical

Table 14. Electrical attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Electrical info			
30701	Electrical type	Type of electrical asset	Choice	
	Motor	Electromechanical device for converting electrical energy into mechanical energy		
	Actuator	Device that uses electrical or pneumatic power to operate valves		
	Power device	Electrical or electronic device for managing the supply and distribution of electrical power		
	Electronic	Component or device that controls the flow of electrical currents for the purpose of information processing and system control		
	Heater	Device that converts electrical energy into heat		
	Blower	Device that uses electrical energy to move or pressurize air		
	Other	Any other type of electrical asset not defined in this section		
30702	Electrical power device type	Classification of electrical power device based on purpose or mode of operation	Choice	Electrical type: Power device
	Control panel	Controls electrical power distribution		
	Circuit breaker	Automatic device for stopping the flow of current in a circuit in case of a fault		
	Uninterruptible power supply (UPS)	Energy storage device that ensures a constant supply of power, even during power outages		
	Transformer	Increases/decreases the		

		voltage/current or visa versa.		
	Power conditioner	Device that improves the quality of power delivered by smoothing out voltage spikes and reducing electrical noise		
	Switch	Device to disconnect or connect the conducting path in an electrical circuit		
	Motor controller	Device for starting, stopping, and regulating the performance and speed of an electric motor		
	Inverter	Device for converting direct current into alternating current		

6.5.3.8 Valve

Table 15. Valve attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Valve info			
30801	Valve type	Purpose or operating principle of a valve	Choice	
	Shutoff	Closes off the source of water or a section of a distribution system		
	Check	Allows flow in only one direction and closes when the flow tries to reverse		
	Backflow preventer	Device for preventing backflow of water into a supply system using a double gate, double check, or air gap assembly		
	Washout	Valve installed in a low point or depression in a pipeline to allow for drainage of the line; also referred to as a blowoff valve		
	Flow control	Pressure-compensating valve installed to regulate the flow of water		
	Pressure reducing	Flow valve used to allow water to flow from higher-pressure side to lower-pressure side so as not to exceed a set maximum pressure on the low-pressure side		
	Pressure sustaining	Valve that opens only as much as necessary to maintain a certain pressure on the inlet side; also		

		referred to as a back pressure regulator		
	Throttle control	Valve installed to reduce the flow through a pipeline		
	Air relief	Air valve placed at the summit of a pipeline to release air automatically to prevent air binding and pressure buildup, or to allow air to enter a line if the internal pressure becomes less than atmospheric pressure		
	Other	Any type of valve not otherwise defined in this section		
30802	Date of last actuation	Most recent date that the valve was actuated to prevent degradation and verify functionality	Date	

6.5.3.9 Hydrant

Table 16. Hydrant attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Hydrant info			
30901	Hydrant type	Type of hydrant design	Choice	
	Wet barrel	Shut-off valves are located above ground, so there is water in the barrel even when not in use		
	Dry barrel	Shut-off valve is located underground and the barrel is drained or pumped dry when not in use to protect from freezing		
	Standpipe	Connections for fire hoses within a building		

6.5.3.10 Junction

Table 17. Junction attributes

Code	**Group**, Attribute, or	Description	Data type	Conditions

	Choice Name			
	Junction info			
31001	Junction type		Choice	
	Union	Device used to connect two pipes		
	Elbow	Fitting that connects two pipes at an angle, usually 90 degrees		
	Tee	Fitting in the shape of the letter T that has an opening perpendicular to the main flow pathway, allowing a three-way connection		
	Sleeve	Fitting for uniting two pipes of the same nominal diameter in a straight line		
	Cap	Fitting to make the end of a pipe watertight		
	Reducer	Fitting used to join two different sized pieces of pipe		
	Cross	Fitting with four branches arranged at right angles to each other		
	Outside threaded adapter	Fitting with threads on the outside of the pipe; previously referred to by the outdated term “male adapter”		
	Inside threaded adapter	Fitting with threads on the inside of the pipe; previously referred to by the outdated term “female adapter”		
	Other	Any type of junction not otherwise described in this section		

6.5.3.11 Sampling point

Table 18. Sampling point attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Sampling point info			
31101	Sampling point type	Type of facility that allows for collection of water quality samples	Choice	
	Faucet	Single device to shut off or control		

		the flow from a pipe		
	Sampling station	Enclosed box or structure with a faucet or valve and spigot inside		
	Sampling port	Fitting or valve attached to a pipe		
	Other	Any type of sampling point not otherwise defined in this section		

6.5.3.12 Sensor

Table 19. Sensor attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Sensor info			
31201	Sensor type		Choice	
	Pressure	Force pushing on a unit area		
	Level	Height or depth of fluid, measured from a known reference point		
	Moisture	Presence or absence of water in contact with the sensor		
	Current	Rate of flow of electric charge through a unit area		
	Voltage	Electrical pressure available to cause a flow of current		
	Position	Distance or angular position from a known reference point along a line or circle		
	pH	Measure of the acidity or alkalinity of a solution		
	ORP	Oxidation-reduction potential, a measure of the state of oxidation in a solution		
	DO	Dissolved oxygen, the concentration of oxygen in aqueous solution		
	Turbidity	Measurement of the scattering or absorption of light in a solution, which is related to the amount of suspended matter		
	EC	Electrical conductivity, a measure of a fluid's capability to pass an electrical current		
	Free chlorine	Concentration of free available chlorine that is not combined with		

		ammonia or other compounds		
	Total chlorine	Total concentration of chlorine present, regardless of the type of chlorine		
	Nitrate	Oxidized form of nitrogen (NO_3^-)		
	Multiparameter	More than one parameter is detected by the sensor		
	Other	Any other parameter not otherwise defined in this section		

6.5.3.13 Analyzer

Table 20. Analyzer attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Analyzer info			
31301	Analyzer type	Short description of the type or purpose of the analyzer	Text	

6.5.3.14 Structure

Table 21. Structure attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Structure info			
31401	Structure type		Choice	
	Building	Any enclosed structure designed to protect from the elements		
	Perimeter	Fence, wall, or other protective structure that encloses a land area		
	Road	Linear pathway that has been improved to allow for travel by vehicles and pedestrians		
	Apron	Smooth impermeable surface constructed around a water point to prevent spilt water from pooling or soaking into the ground		
	Utility vault	Underground room or vault providing access and protection to equipment		
	Other	Any type of structure not otherwise defined in this section		
	Perimeter data			Structure type: Perimeter
31411	Perimeter type	Type of perimeter design	Choice	
	Fence	Barrier constructed of vertical posts connected by horizontal sections of sturdy material		
	Wall	Permanent structure with a continuous surface		
	Other	Any type of perimeter structure not defined elsewhere in this section		
31412	Perimeter total length	Total length of the entire perimeter that is being defined as an asset	Unit: Length	
31413	Perimeter height	Typical or minimum height of the perimeter	Unit: Length	
31414	Perimeter setback distance	Typical or minimum distance from the perimeter from the structure or equipment it is designed to protect	Unit: Length	

6.5.3.15 Water point

Table 22. Water point attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Water point info			
31501	Water point type	Type of technology used to access water	Choice	
	Piped connection	Piped water supply connected to a dwelling, building, yard, or plot		
	Kiosk	Water point from which water is sold in small quantities to nearby households		
	Public tap	Public water point from which people can collect water; also known as a standpipe or fountain		
	Dug well	Large-diameter structure for accessing groundwater that is usually dug by hand and lined to prevent collapse; sometimes referred to as a shallow well		
	Spring	Concentrated discharge of groundwater at a location on the surface that is usually protected by a spring box to allow water to flow through a pipe or outlet		
	Borehole	Small-diameter deep hole that has been driven, bored, or drilled in order to reach groundwater		
	Tubewell	Small-diameter hole that is usually hand-driven into a shallow groundwater aquifer; the more general category “borehole” should be used except in areas where tubewells are common		
	Rainwater catchment	System that captures and stores rainwater		
	Watering trough	Receptacle that provides drinking water to animals		
	Other	Any other type of water point not otherwise described in this section		
31502	Agricultural use	Water point provides water that is primarily for agricultural use	Checkbox	
31503	Livestock use	Water point provides water that is primarily for use by livestock	Checkbox	
31504	Meter installed	Meter is installed at this water point	Checkbox	

31505	Number of taps	Total number of taps, such as faucets or spigots, available at a kiosk or public tap	Number	Water point type: Kiosk, Public tap
31506	Kiosk tank installed	Presence of a tank at the kiosk for storing water during planned or unplanned outages	Checkbox	Water point type: Kiosk
31507	Connection type	Categorization of piped connections based on the type of user or customer	Choice	Water point type: Piped connection
	Residential	Water provided to households for domestic use in the home and on the premises		
	Commercial	Water provided to a business for commercial use or resale		
	Industrial	Water used by industry for fabrication, processing, washing, and cooling		
	Institutional	Water provided to facilities dedicated to public service, including schools, healthcare facilities, courthouses, markets, and government facilities		
	Other	Any other type of water connection not otherwise defined in this section		
31508	Connection diameter	The diameter of pipe used for the customer connection	Unit: Length	Water point type: Piped connection

6.5.3.16 Other vertical

Table 23. Other vertical attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Other vertical info			
39901	Other vertical asset type	Description or categorization of the other type of vertical asset	Text	

6.5.4 Horizontal types

6.5.4.1 Pipe

Table 24. Pipe attributes

Code	**Group**, Attribute, or *Choice* Name	Description	Data type	Conditions
	Pipe info			
40101	Pipe type	Category of water supply pipe type	Choice	
	Transmission main	Used for transporting water from the source of supply or treatment to storage reservoirs or centralized points of distribution		
	Primary main	Large main that transports water from storage reservoirs or centralized points of distribution to various points of connection with the distribution grid		
	Secondary main	Smaller feeder main that connects from primary main to distribution mains that provide water to consumers		
	Distribution main	Water main that is located near water users and provides water to consumers either directly or through connections with service lines; sometimes referred to as a tertiary main		
	Service line	Small diameter pipe that connects directly to water consumers, often through a service valve and meter		
	Other	Any other type of water main not defined in this section		
40102	Pipe color	Color or pattern painted on pipe to designate the type of fluid	Text	
40103	Nominal pressure	Numerical designation of design pressure that is a convenient round number for reference purposes	Unit: Pressure	
40104	Pipe depth below ground	Depth of the pipe below the local ground surface	Unit: Length	

6.5.4.2 Canal

This section is reserved.

6.5.5 Natural types

This section is reserved.

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